

A common neighbourhood geometry for reference evidence and background

Implementation derivation: experimental single-chain matched-PSSM score

Populations. A query q is a full junction including conserved anchors. R_X contains compatible reference junction representatives for epitope X ; B contains unique full control junctions. Set $N_X = |R_X|$, $M = |B|$. References retain the first source-order V call per junction. Exact junction identity is excluded symmetrically, while population denominators are retained; unusable references remain in N_X and are reported separately. Controls are the declared generation/repertoire population, not other VDJdb epitopes. No assumption of naive-cell provenance follows from the score.

Geometry. Let $s(a, b)$ be BLOSUM62 and $\Delta(a, b) = s(a, a) + s(b, b) - 2s(a, b)$ its native nonnegative integer penalty. Use $w_{L,j} = \max\{1, \text{round}(100\omega_{L,j})\}$, where ω is the shipped end-anchored empirical significance profile, mean one. For aligned residue columns $A(q, r)$, $L = \max(\ell_q, \ell_r)$ and $h = |\ell_q - \ell_r|$,

$$d(q, r) = \sum_{(i,j) \in A(q,r)} w_{L,\max(i,j)} \Delta(q_i, r_j) + g(h), \quad g(0) = 0, \quad g(h) = 2800 + 1400(h - 1) \quad (h > 0). \quad (1)$$

The single length-difference block lies after six matched prefix residues, clamped to the shorter junction length (Cys index zero). Equal lengths have no gap. This is fixed-gap, not general local alignment. The cutoff $D = 7000$ excludes $h > 4$. The separate historical V factor $a(q, r)$ is one for identical allele-stripped genes, otherwise 0.25 vsim(V_q, V_r). It is not a V-loop likelihood or a CDR1/CDR2/CDR2.5 distance.

One score. Define the nonexact empirical background ball mass and exponential kernel:

$$\hat{F}_q(t) = \frac{1}{M} \sum_{b \in B} \mathbf{1}\{b \neq q, d(q, b) \leq t\}, \quad K_\tau(t) = e^{-t/\tau} \mathbf{1}\{t \leq D\}. \quad (2)$$

The implemented statistic is

$$S_X(q; \tau) = \sum_{r \in R_X, r \neq q} \frac{a(q, r) K_\tau(d(q, r))}{\max\{N_X \hat{F}_q(d(q, r)), \varepsilon\}}, \quad \varepsilon = 0.01, \quad \tau = 400 \text{ by default}. \quad (3)$$

Writing $n_X^a(t) = \sum_{r \neq q} a(q, r) \mathbf{1}\{d(q, r) \leq t\}$ gives the compact form

$$S_X = \int_{[0,D]} \frac{e^{-t/\tau}}{\max\{N_X \hat{F}_q(t), \varepsilon\}} dn_X^a(t). \quad (4)$$

Thus close reference evidence is weighted by the expected unweighted count under the identical neighbourhood predicate. No length switch or top- K cap is needed in this expression. A single-distance shell with unit V weights reduces to $e^{-t/\tau} n_X(t) / \max\{N_X \hat{F}_q(t), \varepsilon\}$.

Derivation and statistical meaning. For independent reference draws from the control distribution, a fixed-ball count has mean $N_X F_q(t)$ and binomial variance $N_X F_q(t)(1 - F_q(t))$. Rare balls allow a Poisson approximation. This establishes the expected-count denominator, not a P-value for the sum. V factors occur only in the numerator: the denominator is not an expected V-weighted count. Finite-control uncertainty is not integrated into S_X . With known F_q and unit V weights,

$$\mathbb{E}_0[S_X] = N_X \int_{[0,D]} \frac{e^{-t/\tau}}{\max\{N_X F_q(t), \varepsilon\}} dF_q(t) \neq 1 \quad \text{in general}. \quad (5)$$

Without the floor, the continuous part is $\int e^{-t/\tau} d \log F_q(t)$, potentially divergent near zero mass. The floor bounds each contribution by $a(q, r)/\varepsilon$. It is inherited regularisation, not exact finite-control calibration.

Historical correspondence and limiting cases

Original formula. The original equal-length, at-most-five-substitution density is

$$S_X^{\text{old}}(q) = \sum_{\substack{r \neq q, \ell_r = \ell_q \\ h(q,r) \leq 5}} \frac{a(q,r)e^{-p(q,r)/400}}{\max\{N_X \widehat{F}_q^{\text{edit}}(h(q,r)), 0.01\}}. \quad (6)$$

Here $p(q,r)$ is the positional BLOSUM penalty and $h(q,r)$ the substitution count. The denominator counts the ordinary unit-edit ball, allowing the declared historical insertion/deletion scope for controls. The numerator and denominator therefore use different geometries. This is deliberate reproduction of the earlier implementation.

Matched mode uses the same positional distance in the kernel and background CDF and rescores every accepted nonexact neighbour within D . It does not add its score to the old density. The old and new scores agree only when accepted edges, penalties, V factors and background masses agree. Equal length alone does not imply this equality. The separate additive historical-PSSM experiment retains the old density and adds only edges outside the original substitution ball.

Full output score. When the original germline prior is requested, its fusion is preserved:

$$T_X(q) = \begin{cases} \log(S_X(q) + \eta) + G_X(q), & N_X < 500, \\ S_X(q), & N_X \geq 500, \end{cases} \quad \eta = 10^{-6}. \quad (7)$$

Without that option, the output is S_X . The original prior is a sum over features f (V/J/length for beta, V/J for alpha) with category counts $c_{X,f}, c_{B_G,f}$ and raw prior-background size M_G :

$$G_X(q) = \sum_f \log \left[\frac{(c_{X,f}(f(q)) + 0.5)/(N_X + 30)}{(c_{B_G,f}(f(q)) + 0.5)/(M_G + 30)} \right]. \quad (8)$$

This prior population B_G is separate from the unique-junction control B ; raw multiplicities are retained. The constants and 500-reference switch are historical conventions, not consequences of the kernel derivation. Original alpha reproduction deliberately uses the declared original beta prior background. Neither that choice nor paired cohort-rank fusion defines a new probabilistic model.

Limiting cases.

- No accepted reference neighbours gives $S_X = 0$.
- As $\tau \rightarrow 0$, positive-penalty edges vanish; nonexact zero-penalty edges, if present, retain their contribution. For positive nonexact penalties, the sparse-reference output tends to $\log \eta + G_X$: changing τ changes density/prior balance, so better density ranking need not improve fused ranking.
- As $\tau \rightarrow \infty$, attenuation disappears within D : this remains a sum of inverse-background-mass contributions, not one ball-count ratio.
- Uniform position weights and forbidden gaps give weighted-BLOSUM Hamming geometry; unit substitution penalties give ordinary Hamming geometry.
- More controls at fixed distribution stabilise $\widehat{F}_q$; they do not multiply the score by M .
- Integer penalties, ties, neighbour-set changes and the cutoff remain discrete. Exponential weighting does not prove global smoothness.

Verification boundary. The shipping CLI exposes half/default/double scales $\tau = 200, 400, 800$ while holding geometry, cutoff, controls and the original comparator fixed. These are diagnostic choices, not a fitted optimum or a superiority claim. The existing `p_enrichment` is the original radius-one component test; it does not calibrate S_X or T_X . The present derivation is single-chain. Stable paired joint scoring, joint calibration and integration of V-loop distances remain separate work. M3 is outside this derivation.